TOPIC LOOKUP — READ FIRST

average power density	POWER R&T (1,1)
angle of reflection	ANGLES (1,2)
critical angle	ANGLES (1,2)
angle of refraction	ANGLES (1,2)
angle of transmission	ANGLES (1,2)
tangent of Brewster angle	ANGLES (1,2)
Brewster angle	ANGLES (1,2)
polarization for reflected/transmitted waves	ANGLES coral note (1,2)
enclosing a waveguide on both sides w/ length	CAVITY RES. (1,3)
large aperture	APERTURE ANT. (2,3)
circular aperture	APERTURE ANT. HPBW tbl. (2,3)

$\tilde{E}$ PHASOR TEMPLATES (Air: $\varepsilon_r=\mu_r=1$)

Wavenumber per medium *nonmag.*

$$k_i=\frac{\omega\sqrt{\varepsilon_{r,i}}}{c}\ \text{(rad/m)}\quad\Rightarrow\ k_2=k_1\sqrt{\varepsilon_{r,2}/\varepsilon_{r,1}}$$

Direction table *$u\in\{x,y,z\}$ = axis perp. to boundary*

Inc dir	Inc	Refl	Trans
$+\hat{u}$	e^{-jk_1u}	e^{+jk_1u}	e^{-jk_2u}
$-\hat{u}$	e^{+jk_1u}	e^{-jk_1u}	e^{+jk_2u}

$+\hat{u}$ direction $\Leftrightarrow$ Med 2 at $u\geq0$. Reflected sign flips, transmitted matches incident.

General template (incident has pol. $\hat{e}$, amplitude a_0):

$$\tilde{\mathbf{E}}^i=a_0\hat{e}\,e^{\mp jk_1u}\ \text{(V/m)}\qquad\text{(incident wave, Med 1)}$$

$$\tilde{\mathbf{E}}^r=\Gamma\,a_0\hat{e}\,e^{\pm jk_1u}\ \text{(V/m)}\qquad\text{(reflected wave, Med 1)}$$

$$\tilde{\mathbf{E}}^t=\tau\,a_0\hat{e}\,e^{\mp jk_2u}\ \text{(V/m)}\qquad\text{(transmitted wave, Med 2)}$$

(lossy med 2: replace $e^{\mp jk_2u}$ with $e^{\mp\alpha_2u}e^{\mp j\beta_2u}$)

Total in medium 1: $\tilde{\mathbf{E}}_1=\tilde{\mathbf{E}}^i+\tilde{\mathbf{E}}^r$.

$\hat{e}$ by polarization *plug into templates above*

Pol.	$\hat{e}$	Note
Lin. $\hat{x}$	$\hat{x}$	$E_y=0$
Lin. $\hat{y}$	$\hat{y}$	$E_x=0$
Lin. at 45°	$(\hat{x}+\hat{y})/\sqrt{2}$	in phase
RHC	$\hat{x}-j\hat{y}$	$\delta=-\pi/2$
LHC	$\hat{x}+j\hat{y}$	$\delta=+\pi/2$
General	$a_x\hat{x}+a_ye^{j\delta}\hat{y}$	elliptical

Γ,τ act on $\hat{e}$ unchanged. Only swap the polarization, not the formulas.

TABLE 8-2 — Γ,τ,R,T (unitless) SUMMARY

***Med. 1** ($z<0$) incident side; **Med. 2** ($z\geq0$) transmitted.*

	Normal ($\theta_i=0$)	**$\perp$ pol**	**$\parallel$ pol**
Γ refl. wav	$\frac{\eta_2-\eta_1}{\eta_2+\eta_1}$	$\frac{\eta_2\cos\theta_i-\eta_1\cos\theta_t}{\eta_2\cos\theta_i+\eta_1\cos\theta_t}$	$\frac{\eta_2\cos\theta_t-\eta_1\cos\theta_i}{\eta_2\cos\theta_t+\eta_1\cos\theta_i}$
τ trans. wav	$\frac{2\eta_2}{\eta_2+\eta_1}$	$\frac{2\eta_2\cos\theta_i}{\eta_2\cos\theta_i+\eta_1\cos\theta_t}$	$\frac{2\eta_2\cos\theta_i}{\eta_2\cos\theta_t+\eta_1\cos\theta_i}$
τ rel.	$\tau=1+\Gamma$	$\tau_\perp=1+\Gamma_\perp$	$\tau_\parallel=(1+\Gamma_\parallel)\frac{\cos\theta_i}{\cos\theta_t}$
R refl. pwr	$\lvert\Gamma\rvert^2$	$\lvert\Gamma_\perp\rvert^2$	$\lvert\Gamma_\parallel\rvert^2$
T trans. pwr	$\lvert\tau\rvert^2\frac{\eta_1}{\eta_2}$	$\lvert\tau_\perp\rvert^2\frac{\eta_1\cos\theta_t}{\eta_2\cos\theta_i}$	$\lvert\tau_\parallel\rvert^2\frac{\eta_1\cos\theta_t}{\eta_2\cos\theta_i}$
$R+T$ total	$T=1-R$	$T_\perp=1-R_\perp$	$T_\parallel=1-R_\parallel$

Notes: $\sin\theta_t=\sqrt{\mu_1\varepsilon_1/\mu_2\varepsilon_2}\sin\theta_i$; *nonmag.* $\Rightarrow\eta_2/\eta_1=n_1/n_2$.

Intrinsic impedance η_i *plug into table above*

$$\eta_i=\sqrt{\frac{\mu_r}{\varepsilon_r}}\,\eta_0\xrightarrow{\mu_r=1}\frac{\eta_0}{\sqrt{\varepsilon_{r,i}}}\ (\Omega)\qquad\eta_0=120\pi\approx377\,\Omega$$

Default $\mu_r=1$ (air, dielectric). Use $\mu_r\neq1$ only if problem says "ferromagnetic"/iron/ferrite/etc.

Low-loss correction η_c $\sigma/(\omega\varepsilon)\ll1$

$$\eta_c\approx\sqrt{\frac{\mu}{\varepsilon'}}\left(1+j\frac{\sigma}{2\omega\varepsilon'}\right)\xrightarrow{\text{drop }j}\frac{\eta_0}{\sqrt{\varepsilon_r}}$$

For Γ/τ just use $\eta_0/\sqrt{\varepsilon_r}$. The j term is for $\lvert\eta_c\rvert,\theta_\eta$. See *LOSSY MEDIA — 5 CASES.*

POWER REFLECTION & TRANSMISSION

Avg power density of plane wave *lossless; $\lvert E_0\rvert$ peak E-field amp.*

$$S_{\text{av}}=\frac{\lvert E_0\rvert^2}{2\eta}\ (\text{W/m}^2)$$

η = intrinsic impedance of the medium (Ω): $\eta=\sqrt{\mu/\varepsilon}=\eta_0\sqrt{\mu_r/\varepsilon_r}\xrightarrow{\mu_r=1}\eta_0/\sqrt{\varepsilon_r}$; $\eta_0=120\pi\approx377\,\Omega$ *(free space).*

% reflected *always* $\%P_r=100\,\lvert\Gamma\rvert^2$

% transmitted *η-ratio matters!* $\%P_t=100\,\lvert\tau\rvert^2\,\eta_1/\eta_2$

Check: $\%P_r+\%P_t=100$.

ANGLES — REFLECTION, TRANSMISSION, CRITICAL, BREWSTER

Reflection angle (Snell, reflection) *always* $\theta_r=\theta_i$ (from normal)

Transmission/refraction (Snell, refraction) *single boundary; angles from normal*

$$\sin\theta_t=\sin\theta_i\sqrt{\frac{\varepsilon_{r1}}{\varepsilon_{r2}}}=\sin\theta_i\,\frac{n_1}{n_2}$$

Multilayer chain (Snell, chained) *stack of slabs $1\to2\to3\to4$*

$$\sin\theta_{i+1}=\sin\theta_i\sqrt{\frac{\varepsilon_{r,i}}{\varepsilon_{r,i+1}}}$$

Air–slabs–air shortcut *outer media identical; collapse of Snell chain*

$$\theta_{\text{exit}}=\theta_{\text{incident}}$$

Critical angle θ_c *TIR when $n_1>n_2$; no transmitted wave*

$$\sin\theta_c=\frac{n_2}{n_1}=\sqrt{\frac{\varepsilon_{r2}}{\varepsilon_{r1}}}\ \Rightarrow\ \theta_c=\arcsin\left(\frac{n_2}{n_1}\right)$$

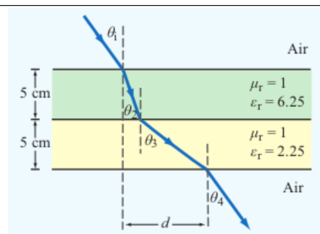
Brewster angle θ_B ($\parallel$-pol only) $\Gamma_\parallel=0$; *nonmag*

$$\tan\theta_B=\frac{n_2}{n_1}=\sqrt{\frac{\varepsilon_{r2}}{\varepsilon_{r1}}}\ \Rightarrow\ \theta_B=\arctan\left(\frac{n_2}{n_1}\right)$$

At θ_B: reflected wave is pure $\perp$-pol ($\parallel$ reflection = 0); transmitted has both.

Snell only sets angles. Reflection/transmission magnitudes need Fresnel coefficients ($\parallel$ vs $\perp$).

LATERAL DISPLACEMENT



Beam offset through N slabs *thickness t_i, refracted angle inside slab i* **Slab-indexed** *θ_i = angle inside slab i; t_i = slab thickness (m)*

$$d=\sum_{i=1}^{N}t_i\,\tan\theta_i$$

The angle (θ_i) in the formula is always the one the ray makes inside that slab of height (t_i).

LOSSY MEDIA — 5 CASES

Loss tangent *$\varepsilon'=\varepsilon_r\varepsilon_0$; $\varepsilon_0=8.854\times10^{-12}$ F/m*

$$\frac{\varepsilon''}{\varepsilon'}=\frac{\sigma}{\omega\varepsilon_r\varepsilon_0}$$

Type	$\sigma/\omega\varepsilon$	$\eta_c,\ \alpha,\ \beta$
Lossless ($\sigma=0$)	0	$\eta=\eta_0/\sqrt{\varepsilon_r}$ exact $\alpha=0$, $\beta=\omega\sqrt{\varepsilon_r}/c$
Low-loss	$<10^{-2}$	$\eta_c\approx\frac{\eta_0}{\sqrt{\varepsilon_r}}\left(1+j\frac{\sigma}{2\omega\varepsilon'}\right)$ $\alpha\approx\frac{\sigma\eta_0}{2\sqrt{\varepsilon_r}}$, $\beta\approx\frac{\omega\sqrt{\varepsilon_r}}{c}$
Quasi-cond.	10^{-2}–10^2	exact: $(\eta/\sqrt{X})e^{j\theta_\eta}$ see below
Good cond.	$>10^2$	$\eta_c\approx(1+j)\sqrt{\pi f\mu/\sigma}$ $\alpha\approx\beta\approx\sqrt{\pi f\mu\sigma}$
Magnetic	any	keep full $\sqrt{\mu_r/\varepsilon_r}\,\eta_0$

For Γ/τ: drop j correction; use $\eta_0/\sqrt{\varepsilon_r}$. Magnetic: keep μ_r.

Low-loss quick params $\sigma/(\omega\varepsilon)\ll1$, $\mu_r=1$

$$\alpha\approx\frac{\sigma\eta_0}{2\sqrt{\varepsilon_r}}\ \text{(Np/m)},\ \beta\approx\frac{\omega\sqrt{\varepsilon_r}}{c}\ \text{(rad/m)},\ \eta_c\approx\frac{\eta_0}{\sqrt{\varepsilon_r}}\ (\Omega)$$

Exact (quasi-cond) $X=\sqrt{1+(\varepsilon''/\varepsilon')^2}$

$$\alpha=\omega\sqrt{\frac{\mu\varepsilon'}{2}}\sqrt{X-1},\quad\beta=\omega\sqrt{\frac{\mu\varepsilon'}{2}}\sqrt{X+1}$$

$$\theta_\eta=\frac{1}{2}\arctan\frac{\sigma}{\omega\varepsilon'}$$

RECTANGULAR WAVEGUIDE — MODES

TE mode $E_z=0,\ H_z\neq0$ **TM mode** $H_z=0,\ E_z\neq0$

Dimensions $a\times b$ with $a>b$ along $\hat{x},\hat{y}$; wave in $\hat{z}$.

E_x **form (TE)** *read m,n from arguments*

$$E_x\propto\cos\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right)\sin(\omega t-\beta z)$$

coef. on $x=m\pi/a$; coef. on $y=n\pi/b$.

Identify TE$_{mn}$ vs TM$_{mn}$ *cos/sin pattern of E_x is identical for both!*

1. Read the problem statement (e.g. "a TE wave" $\to$ TE).
2. If $m=0$ or $n=0$ in the mode label $\Rightarrow$ must be TE (TM forbids 0).
3. Source field $H_z(x,y)$ given $\Rightarrow$ TE. Source field $E_z(x,y)$ given $\Rightarrow$ TM.

TE allows at least one of $m,n\geq1$ (other can be 0). TM needs both $m,n\geq1$. So TE$_{10}$ exists but TM$_{10}$ doesn't.

TABLE 8-3 — FULL FIELD COMPONENTS

$e^{-j\beta z}$ factor omitted. $k_c^2=(m\pi/a)^2+(n\pi/b)^2$. *$\tilde{E}$ in V/m, $\tilde{H}$ in A/m, Z in Ω.*

TE mode *generator: $\tilde{H}_z$*

$$\tilde{H}_z=H_0\cos\frac{m\pi x}{a}\cos\frac{n\pi y}{b}$$

$$\tilde{E}_x=\frac{j\omega\mu}{k_c^2}\left(\frac{n\pi}{b}\right)H_0\cos\frac{m\pi x}{a}\sin\frac{n\pi y}{b}$$

$$\tilde{E}_y=-\frac{j\omega\mu}{k_c^2}\left(\frac{m\pi}{a}\right)H_0\sin\frac{m\pi x}{a}\cos\frac{n\pi y}{b}$$

$$\tilde{E}_z=0,\quad\tilde{H}_x=-\tilde{E}_y/Z_{\text{TE}},\quad\tilde{H}_y=\tilde{E}_x/Z_{\text{TE}}$$

$$Z_{\text{TE}}=\eta/\sqrt{1-(f_c/f)^2}$$

TM mode *generator: $\tilde{E}_z$*

$$\tilde{E}_z=E_0\sin\frac{m\pi x}{a}\sin\frac{n\pi y}{b}$$

$$\tilde{E}_x=-\frac{j\beta}{k_c^2}\left(\frac{m\pi}{a}\right)E_0\cos\frac{m\pi x}{a}\sin\frac{n\pi y}{b}$$

$$\tilde{E}_y=-\frac{j\beta}{k_c^2}\left(\frac{n\pi}{b}\right)E_0\sin\frac{m\pi x}{a}\cos\frac{n\pi y}{b}$$

$$\tilde{H}_z=0,\quad\tilde{H}_x=-\tilde{E}_y/Z_{\text{TM}},\quad\tilde{H}_y=\tilde{E}_x/Z_{\text{TM}}$$

$$Z_{\text{TM}}=\eta\sqrt{1-(f_c/f)^2}$$

TEM plane wave *for reference*

$$\tilde{E}_x=E_0\,e^{-j\beta z},\quad\tilde{H}_y=\tilde{E}_x/\eta$$

$$\eta=\sqrt{\mu/\varepsilon}\ (\Omega),\quad f_c\ \text{N/A},\quad k=\omega\sqrt{\mu\varepsilon}$$

Properties common to TE/TM

$$f_c=\frac{u_{p0}}{2}\sqrt{(m/a)^2+(n/b)^2}\ \text{(Hz)}$$

$$\beta=k\sqrt{1-(f_c/f)^2}\ \text{(rad/m)}$$

$$u_p=\frac{\omega}{\beta}=\frac{u_{p0}}{\sqrt{1-(f_c/f)^2}}\ \text{(m/s)}$$

CUTOFF FREQUENCY

Common cutoffs ($a>b$) *f_c formula in Table 8-3*

Mode	f_{mn}
u_{p0}	$c/\sqrt{\varepsilon_r}$ (hollow: $u_{p0}=c$)
TE$_{10}$	$f_{10}=u_{p0}/(2a)$ (dominant)
TE$_{01}$	$f_{01}=u_{p0}/(2b)$
TE$_{20}$	$f_{20}=u_{p0}/a=2f_{10}$
TE$_{11}$,TM$_{11}$	$f_{11}=\frac{u_{p0}}{2}\sqrt{1/a^2+1/b^2}$

Propagation requires $f>f_{mn}$. In Z_{TE}/Z_{TM} and u_g, $f_c=f_{mn}$ of the excited mode.

CAVITY RESONATOR (guide closed both ends, length d)

Resonant frequencies *m,n,p = mode indices; standing wave in all 3 dims*

$$f_{mnp}=\frac{c}{2\sqrt{\varepsilon_r}}\sqrt{\left(\frac{m}{a}\right)^2+\left(\frac{n}{b}\right)^2+\left(\frac{p}{d}\right)^2}\ \text{(Hz)}$$

TE$_{mnp}$: $p\geq1$, at least one of $m,n\geq1$. TM$_{mnp}$: $m,n\geq1$, $p\geq0$.

Dominant TE$_{101}$ when $a\geq d>b$: $f_{101}=\frac{c}{2\sqrt{\varepsilon_r}}\sqrt{1/a^2+1/d^2}$.

ε_r FROM DISPERSION

Given ω, β, mode (m,n) *result is unitless*

$$\varepsilon_r=\frac{c^2}{\omega^2}\left[\beta^2+\left(\frac{m\pi}{a}\right)^2+\left(\frac{n\pi}{b}\right)^2\right]\ \text{(unitless)}$$

Inputs: ω (rad/s), β (rad/m), a,b (m), $c=3\times10^8$ (m/s), m,n (unitless).

GROUP VELOCITY & TRAVEL TIME

$$u_g=u_{p0}\sqrt{1-(f_{mn}/f)^2}\ \text{(m/s)},\quad u_pu_g=u_{p0}^2$$

$$t=L/u_g\ \text{(s)}\quad\text{travel time through guide of length }L$$

Higher modes have lower u_g $\Rightarrow$ arrive later.

BOUNDARY

η_1,η_2 — imp. (Ω)
Γ,τ,R,T — coeffs (unitless)
$\tau=1+\Gamma$; $R=\lvert\Gamma\rvert^2$
$k_i=\omega\sqrt{\varepsilon_{r,i}}/c$ (rad/m)
α_2 (Np/m), β_2 (rad/m): lossy
$z=0$ — boundary plane (m)
$\eta_0=120\pi\approx377\,\Omega$

SNELL / OBLIQUE

θ_i,θ_t — angles from normal
$n_i=\sqrt{\varepsilon_{r,i}}$ — index of refr.
t_i — slab thickness (m)
d — lateral disp. (m)
plane of inc. = plane of k+normal
$\perp$: $\mathbf{E}\perp$ this plane
$\parallel$: $\mathbf{E}\parallel$ this plane

WAVEGUIDE

a,b — wide, narrow dim. (m); $a>b$
m,n — mode indices (int.)
f_{mn} — cutoff (Hz)
β — prop. const (rad/m)
$Z_{\text{TE/TM}}$ — wave imp. (Ω)
E_x — E-field comp. (V/m)
H_y — H-field comp. (A/m)
$\eta=\eta_0/\sqrt{\varepsilon_r}$

VELOCITIES

$u_{p0}=c/\sqrt{\varepsilon_r}$ — bulk phase vel.
$u_p=u_{p0}/\sqrt{1-(f_c/f)^2}$
$u_g=u_{p0}\sqrt{1-(f_c/f)^2}$
$u_pu_g=u_{p0}^2$
$t=L/u_g$ — travel time
$c=3\times10^8$ m/s

POWER & UNITS

$\%P_r,\%P_t$ — pwr refl./trans. (unitless)
$\%P_r=100\lvert\Gamma\rvert^2$
$\%P_t=100\lvert\tau\rvert^2\eta_1/\eta_2$
$\%P_r+\%P_t=100$
σ (S/m), ε (F/m), μ (H/m)
$\varepsilon_0=8.85\times10^{-12}$ F/m
$\mu_0=4\pi\times10^{-7}$ H/m

DIPOLE TYPE BY LENGTH
l = antenna rod length (m)

$$\boxed{\lambda \text{ (wavelength)} = \frac{c}{f} = \frac{3\times10^8}{f} \text{ (m)}}$$

l/λ	Type	Use formula
$< 1/50$	Hertzian (short)	HERTZIAN $S(R,\theta)$, $D=1.5$
$= 1/4$	Quarter-wave	ARB formula, $\pi l/\lambda = \pi/4$
$= 1/2$	Half-wave	half-wave shortcut, $D=1.64$, $R_{\text{rad}}=73\,\Omega$
$= 1$	Full-wave	ARB, $D=2.41$, $R_{\text{rad}}=199\,\Omega$
other	Arbitrary	ARB formula

HERTZIAN DIPOLE — $S(R,\theta)$

Far-field E,H phasors *small $l \ll \lambda$*

$$\tilde{E}_\theta = \frac{jI_0 k l \eta_0}{4\pi}\frac{e^{-jkR}}{R}\sin\theta, \quad \tilde{H}_\phi = \tilde{E}_\theta/\eta_0$$

Avg power density *$l/\lambda < 1/50$; far field*

$$\boxed{S(R,\theta) = \frac{\eta_0\, k^2 I_0^2\, l^2}{32\pi^2 R^2}\sin^2\theta} \quad (\text{W/m}^2)$$

Equivalent: $S = S_{\max}\sin^2\theta$ with $S_{\max} = \eta_0 k^2 I_0^2 l^2/(32\pi^2 R^2)$ at $\theta=\pi/2$.

l: rod length (m)	I_0: peak current (A)
R: obs. distance (m)	θ: angle from dipole axis (rad)
$k = 2\pi/\lambda$ (rad/m)	$\eta_0 = 120\pi \approx 377\,\Omega$

Total radiated power $P_{\text{rad}} = \frac{\eta_0\pi}{3}(I_0 l/\lambda)^2$ (W)

ARBITRARY-LENGTH DIPOLE — $S(\theta)$

Avg power density *any l; far field*

$$\boxed{S(R,\theta) = \frac{15 I_0^2}{\pi R^2}\left[\frac{\cos\left(\frac{\pi l}{\lambda}\cos\theta\right) - \cos\left(\frac{\pi l}{\lambda}\right)}{\sin\theta}\right]^2} \quad (\text{W/m}^2)$$

At $\theta = 0, \pi$: bracket is 0/0. L'Hôpital $\Rightarrow$ $S=0$ (no axial radiation). TI-36X alt: evaluate bracket at $\theta = 0.001$ rad $\to$ tiny $\to$ 0.

Normalized pattern *dimensionless* $F(\theta) = S(\theta)/S_{\max}$

COMMON ANGLES (deg $\leftrightarrow$ rad, trig values)

Conversion *multiply by $\pi/180$ or $180/\pi$*

$$\theta_{\text{rad}} = \theta_{\text{deg}}\cdot\frac{\pi}{180} \qquad \theta_{\text{deg}} = \theta_{\text{rad}}\cdot\frac{180}{\pi}$$

$^\circ$	rad	$\sin\theta$	$\cos\theta$	$\sin^2\theta$	$\cos^2\theta$
0	0	0	1	0	1
30	$\pi/6$	$1/2$	$\sqrt{3}/2$	$1/4$	$3/4$
45	$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	$1/2$	$1/2$
60	$\pi/3$	$\sqrt{3}/2$	$1/2$	$3/4$	$1/4$
90	$\pi/2$	1	0	1	0
180	π	0	-1	0	1

Antenna formulas usually take θ in rad. RAD mode on calc.

LINEAR ANTENNA ARRAY
N = # of elements, d = interelement spacing (m)

Array factor (uniform, broadside) *$k = 2\pi/\lambda$, $\gamma = kd\cos\theta$*

$$F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}; \quad F_a^{\max} = N^2 \text{ at } \theta=\pi/2$$

Normalize $F_a^{\text{norm}} = F_a/N^2$

HPBW (broadside, uniform) *use Nd, not $(N-1)d$*

$$\beta \approx 0.886\,\frac{\lambda}{Nd}\text{ (rad)} = 50.8^\circ\cdot\frac{\lambda}{Nd}$$

Electronic steering (scan to θ_0) *replace γ with $\gamma' = kd(\cos\theta - \cos\theta_0)$*

$$\delta = \frac{2\pi d}{\lambda}\cos\theta_0\ \left(\frac{\text{rad}}{\text{elt}}\right); \quad \theta_0 = \frac{\pi}{2}\text{:broadside, 0:end-fire}$$

Frequency scan *feed-line incr. $\Delta\ell = n_0\lambda_0$*

$$\cos\theta_0 \approx \frac{n_0\lambda_0}{d}\ \frac{\Delta f}{f_0}$$

Grating lobes appear if $d > \lambda$. Avoid.

BEAM PARAMETERS — β, Ω_p, D

Normalize *always start here*

$$F(\theta) = S(\theta)/S_{\max} \text{ (unitless)}$$

Beamwidth β at threshold X

Set $F(\theta_X) = X$, solve for θ_X; $\beta = 2\theta_X$ (symmetric)

Threshold	$X = 10^{\text{dB}/10}$	θ_X
HPBW (-3 dB)	0.5	$\sqrt{\frac{\ln 2}{a}}$
-6 dB	0.25	$\sqrt{\frac{\ln 4}{a}}$
-10 dB	0.1	$\sqrt{\frac{\ln 10}{a}}$
any $F(\theta_X)$	any X	$F(\theta_X) = X$

Pattern solid angle Ω_p *recognize pattern, look up*

$$\Omega_p = \int_0^{2\pi}\int_0^{\pi} F(\theta)\sin\theta\, d\theta\, d\phi \quad (\text{sr})$$

No ϕ dep.: $\Omega_p = 2\pi\int_0^\pi F(\theta)\sin\theta\, d\theta$.

θ is the integration variable — DON'T plug in a number for θ (e.g. $\theta_{1/2}$). Integrate over $\theta \in [0,\pi]$.

Pattern $F(\theta)$	Ω_p (sr)
Isotropic ($F=1$)	$4\pi \approx 12.57$
Hertzian $\sin^2\theta$	$8\pi/3 \approx 8.38$
Half-wave dipole	≈ 7.66
Narrow Gauss $e^{-a\theta^2}$	$\pi/a = \pi\theta_{1/2}^2/\ln 2$
2-axis HPBW (aperture)	$\beta_{xz}\cdot\beta_{yz}$
Other (use integral)	numerical

TI-36X: `[2nd][d/dx]` for $\int_\square^\square$, RAD mode, multiply by 2π.

Directivity D *always $D = 4\pi/\Omega_p$; common shortcuts below*

$$D = \frac{4\pi}{\Omega_p} \quad \text{(unitless)} \qquad D_{\text{dB}} = 10\log_{10} D$$

WORD TRAP: "direction" = $\theta_{\max}$ (rad); "directivity" = D (unitless).

Gain (if asked) *ξ = radiation efficiency (unitless), $0 < \xi \le 1$*

$$G = \xi D \text{ (unitless)} \qquad G_{\text{dB}} = 10\log_{10} G \text{ (dB)}$$

Lossless / ideal antenna $\Rightarrow$ $G = D$.

$\xi = \frac{R_{rad}}{R_{rad}+R_{loss}} = \frac{P_{rad}}{P_{rad}+P_{loss}}$; $R_{rad} = 2P_{rad}/I_0^2$ (Ω); $R_{in} = R_{rad}+R_{loss}$.

COMMON DIPOLE PARAMETERS

Lossless ($\xi=1$): $G=D$.

Type	l	D	D_{dB}	R_{rad} (Ω)
Isotropic	N/A	1	0	—
Hertzian	$< \lambda/50$	1.5	1.76	$80\pi^2(l/\lambda)^2$
Half-wave	$\lambda/2$	1.64	2.15	73.2
Monopole	$\lambda/4$ (gnd)	1.64	2.15	36.6
Full-wave	λ	2.41	3.82	199

Half-wave: $P_{rad} = 36.6 I_0^2$ W. Monopole ($\lambda/4$ above ground): same patt. as half-wave but P_{rad}, R_{rad} halved.

ANTENNA AREAS (A_e, A_p)

Effective area *antenna's RX cross section; D from BEAM PARAMS or table above*

$$A_e = \frac{\lambda^2 D}{4\pi} \quad (\text{m}^2)$$

Physical cross section *wire dipole, diameter d; l see table above*

$$A_p = l\,d \quad (\text{m}^2\text{, any dipole})$$

Inverse: D from A_e

$$D = \frac{4\pi A_e}{\lambda^2} \quad \text{(unitless)}$$

Thin wire: $A_e \gg A_p$ (e.g., Hertzian $A_e = 3\lambda^2/8\pi \approx 0.119\lambda^2$).

FRIIS TRANSMISSION FORMULA

For each antenna's G: if given by NAME ("half-wave dipole" etc.) $\to$ COMMON DIPOLE PARAMS ($G=D$, lossless). Else (dB/linear value given) $\to$ use as given (convert dB $\to$ linear).

Pwr density at RX, then RX pwr *G_t, G_r linear gains; $\lambda = c/f$*

$$S_r = \frac{G_t P_t}{4\pi R^2}\ (\text{W/m}^2); \quad \boxed{P_r = G_t G_r\left(\frac{\lambda}{4\pi R}\right)^2 P_t}\ (\text{W})$$

Equivalent: $P_r = S_r A_{e,r}$.

Solve for R *range*

$$R = \frac{\lambda}{4\pi}\sqrt{\frac{G_t G_r P_t}{P_r}} \quad (\text{m})$$

Equivalent form using A_e *recv. area form*

$$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}, \quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$$

Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, so $P_r = G_t G_r P_t/L_{fs}$.

General form (off-axis) *when patterns NOT peak-aligned*

$$P_r = G_t G_r\left(\frac{\lambda}{4\pi R}\right)^2 P_t\, F_t(\theta_t,\phi_t) F_r(\theta_r,\phi_r)$$

F_t, F_r normalized patterns at the link directions; $F = 1$ when aligned.

Recipe 1. $\lambda = c/f$. 2. Convert any dB gain to linear: $G = 10^{G_{\text{dB}}/10}$.
3. Get G_t, G_r by antenna type:
- named dipole $\to$ COMMON DIPOLE PARAMS ($G = D$ lossless)
- aperture / dish $\to$ APERTURE ANTENNAS ($G = D \approx 4\pi A_p/\lambda^2$)
- arbitrary pattern $F(\theta)$ $\to$ BEAM PARAMS ($G = \xi\cdot 4\pi/\Omega_p$)
- given in dB $\to$ dB$\leftrightarrow$LINEAR table

4. Plug into Friis.

Use linear G, not dB. Convert FIRST.

dB $\leftrightarrow$ LINEAR

Works for any power-like quantity (G, D, P_r/P_t, SNR, ...):

$$X_{\text{dB}} = 10\log_{10} X_{\text{lin}} \qquad X_{\text{lin}} = 10^{X_{\text{dB}}/10}$$

Memorize: 3 dB $\Leftrightarrow$ $\times 2$, 10 dB $\Leftrightarrow$ $\times 10$. Add dB = multiply linear.

For E-field / amplitude use 20 log not 10 log (power $\propto E^2$).

NOISE & SNR

Noise power *thermal noise into matched receiver*

$$P_N = k_B\, T_{\text{sys}}\, B \text{ (W)}$$

$k_B = 1.38\times10^{-23}$ J/K; T_{sys} system noise temp (K); B receiver bandwidth (Hz).

Signal-to-noise ratio

$$\text{SNR} = \frac{P_{\text{rec}}}{P_N} \text{ (unitless)} \qquad \text{SNR}_{\text{dB}} = 10\log_{10}\frac{P_{\text{rec}}}{P_N}$$

APERTURE ANTENNAS (horn, dish; large aperture, $d \ge$ several λ)

A_p physical aperture area (m^2); A_e effective area (m^2, $= \lambda^2 D/4\pi$); β_{xz}, β_{yz} HPBWs in two planes (rad); ℓ, d aperture side / diameter (m).

Far-field req.: $R \ge 2d^2/\lambda$ (d = largest aperture dim.).

Pattern (rect., uniform illum.) *peak at $\theta = 0$ (broadside)*

$$F(\theta) = \text{sinc}^2\left(\frac{\pi\ell_x\sin\theta}{\lambda}\right); \quad S_0 = \frac{E_0^2 A_p^2}{2\eta_0\lambda^2 R^2}$$

Directivity (any shape, uniform illum.)

$$D \approx \frac{4\pi A_p}{\lambda^2} \text{ (unitless)} \quad \Rightarrow \quad A_e \approx A_p$$

Directivity from beamwidths *any aperture*

$$D \approx \frac{4\pi}{\beta_{xz}\beta_{yz}} \text{ (use rad); circular: } D \approx 4\pi/\beta^2$$

HPBW and D by aperture shape *uniform illum.; β in rad*

Shape	A_p	HPBW (rad)	D (unitless)
Rect. $\ell_x \times \ell_y$	$\ell_x\ell_y$	$\beta_x \approx 0.88\lambda/\ell_x$ $\beta_y \approx 0.88\lambda/\ell_y$	$D \approx 4\pi/(\beta_x\beta_y)$ $= 4\pi\ell_x\ell_y/\lambda^2$
Square ℓ	ℓ^2	$\beta \approx 0.88\lambda/\ell$	$D \approx 4\pi/\beta^2 = 4\pi\ell^2/\lambda^2$
Circular (diam. d)	$\pi d^2/4$	$\beta \approx 1.02\lambda/d$	$D \approx 4\pi/\beta^2 = \pi^2 d^2/\lambda^2$

Scaling (any ratio) *$D \propto A_p/\lambda^2$; $\beta \propto \lambda/\sqrt{A_p}$*

$$D_{\text{new}} = D_{\text{old}}\cdot\frac{A_{p,\text{new}}}{A_{p,\text{old}}}\cdot\left(\frac{f_{\text{new}}}{f_{\text{old}}}\right)^2$$

$$\beta_{\text{new}} = \beta_{\text{old}}\cdot\sqrt{\frac{A_{p,\text{old}}}{A_{p,\text{new}}}}\cdot\frac{f_{\text{old}}}{f_{\text{new}}}$$

If a parameter is unchanged, its ratio = 1. Examples: double area $\to$ $D\times 2$, $\beta\times 1/\sqrt{2}$; double freq $\to$ $D\times 4$, $\beta \times 1/2$.

DIPOLE

l — antenna length (m)
$\lambda = c/f$ (m); $c = 3\times10^8$ m/s
I_0 — peak current (A)
$k = 2\pi/\lambda$ (rad/m)
R — obs. dist. (m); θ from axis
$\eta_0 = 120\pi \approx 377\,\Omega$
$S(R,\theta)$ — pwr density (W/m^2)

BEAM

$F(\theta) = S/S_{\max}$ (unitless)
a — coef. on θ^2 in Gaussian
β — HPBW (rad or $^\circ$)
Ω_p — pattern solid angle (sr)
$D = 4\pi/\Omega_p$ — directivity
ξ radiation eff. (unitless); $G = \xi D$
$D_{\text{dB}} = 10\log_{10} D$; $D = 10^{D_{\text{dB}}/10}$

FRIIS / dB

P_t — power transmitted (W)
P_r — power received (W)
G_t — transmit gain (linear)
G_r — receive gain (linear)
$S_r = G_t P_t/(4\pi R^2)$ (W/m^2)
$P_r = G_t G_r(\lambda/4\pi R)^2 P_t$
$G = 10^{G_{\text{dB}}/10}$
3 dB $= \times 2$, 10 dB $= \times 10$
$P_N = k_B T_{\text{sys}} B$ (W)

APERTURE

A_p — physical area (m^2)
$A_e = \lambda^2 D/(4\pi)$ (m^2)
$A_e \approx A_p$ for aperture
$D \approx 4\pi A_p/\lambda^2$ (unitless)
$\beta \approx 0.88\lambda/\ell$ (rect/sq.)
$\beta \approx 1.02\lambda/d$ (circular)
$2A_p \Rightarrow 2D$, $\beta/\sqrt{2}$

ARRAY

N — number of elements
d — spacing (m)
$\gamma = kd\cos\theta$
$F_a = \sin^2(N\gamma/2)/\sin^2(\gamma/2)$
$F_a^{\max} = N^2$ at $\theta = \pi/2$
$F_a^{\text{norm}} = F_a/N^2$
$\beta \approx 0.886\lambda/(Nd)$ (broadside)